哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

《设计与制造》

第7章　标准件及常用件

概念

标准件：已完全标准化，由专业工厂生产的零部件

螺纹紧固件、键、销、滚动轴承等

常用件：

齿轮、涡轮、轴、弹簧等

一、螺纹的基本知识

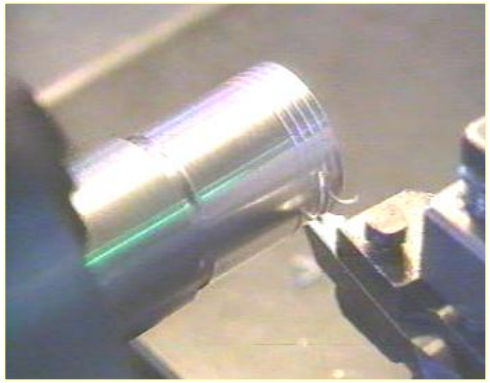
1、螺旋线的形成

2、螺纹的形成及加工方法

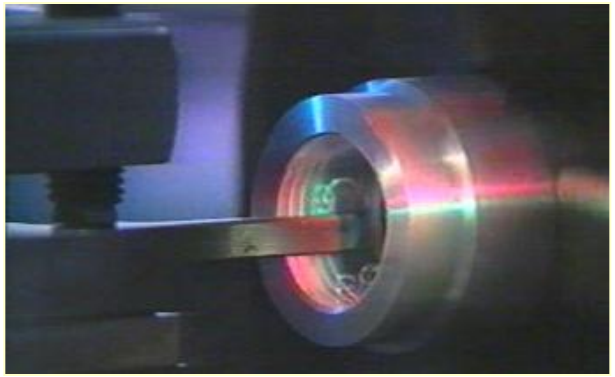
工件绕轴线作等速圆周运动，刀具沿轴线作直线运动且切入工件一定深度

（1）车制螺纹

车制外螺纹



车制内螺纹



需车刀伸入圆柱孔内

（2）手工制作螺纹

丝锥 →内螺纹

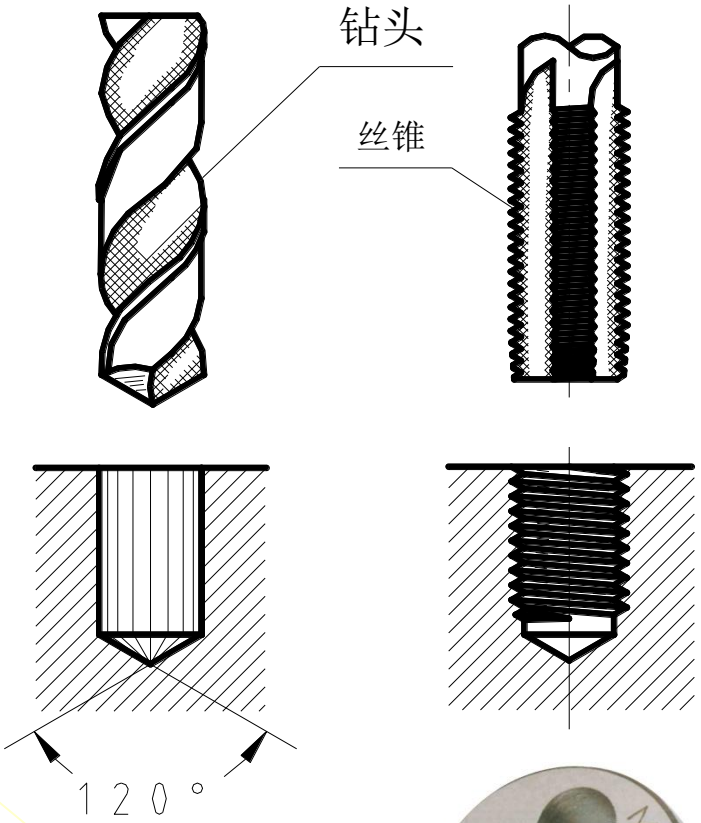
板牙 →外螺纹

丝锥攻丝的方法：

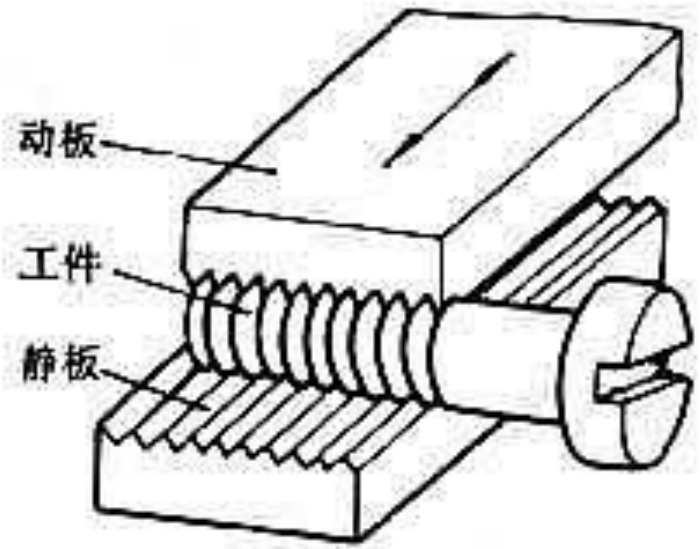
1. 钻头钻孔
2. 丝锥攻丝

1.钻孔深度>螺纹长度

2.钻头锥角



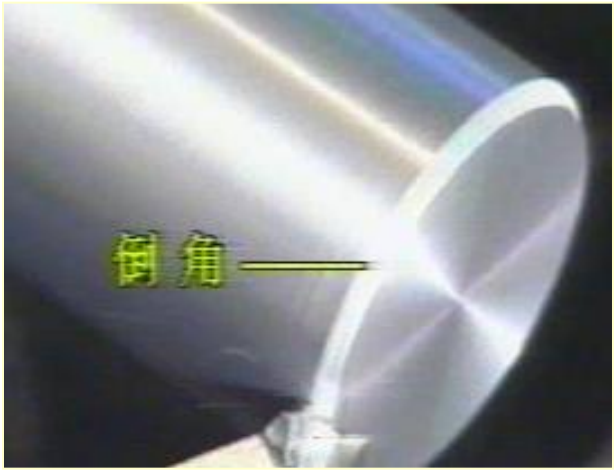
（3）滚压螺纹



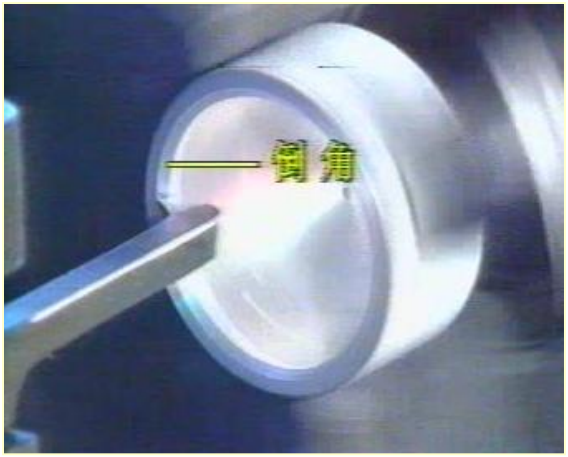
二、螺纹的工艺结构

（1）倒角

为了便于装配、防止螺纹端部损坏，通常在螺纹的起始端，车90度锥面，称倒角。



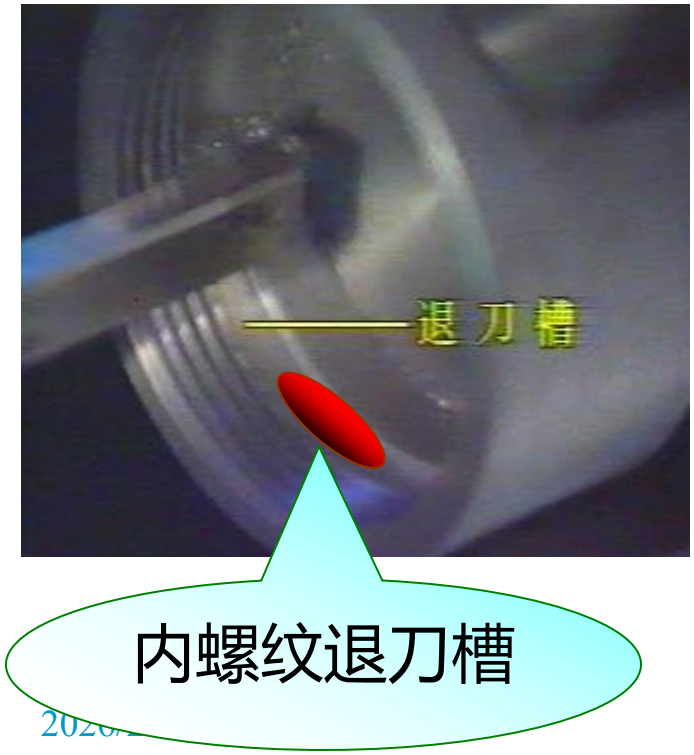
外螺纹倒角

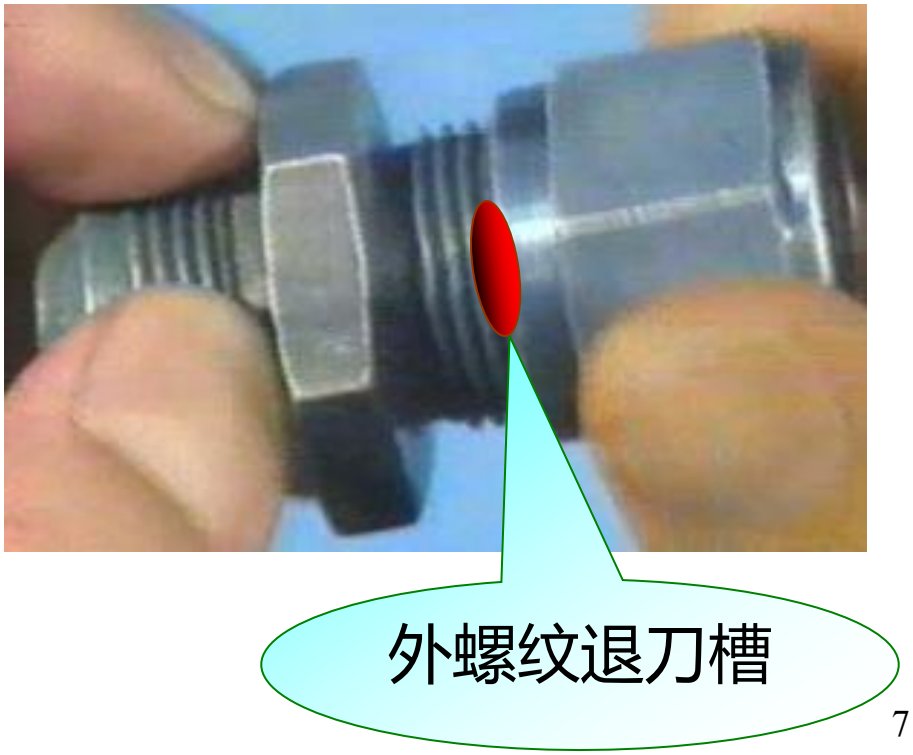


内螺纹倒角

(2) 退刀槽

在螺纹终止处切削一个槽，以便于刀具退出，同时避免出现螺尾。





三、螺纹的结构要素

（1）牙型

（2）公称直径

（3）线数

（4）螺距和导程

（5）旋向

牙型、公称直径和螺距均符合国标 →标准螺纹

牙型符合，其它不符合国标 →特殊螺纹

牙型不符合国标 →非标螺纹

（1）牙型

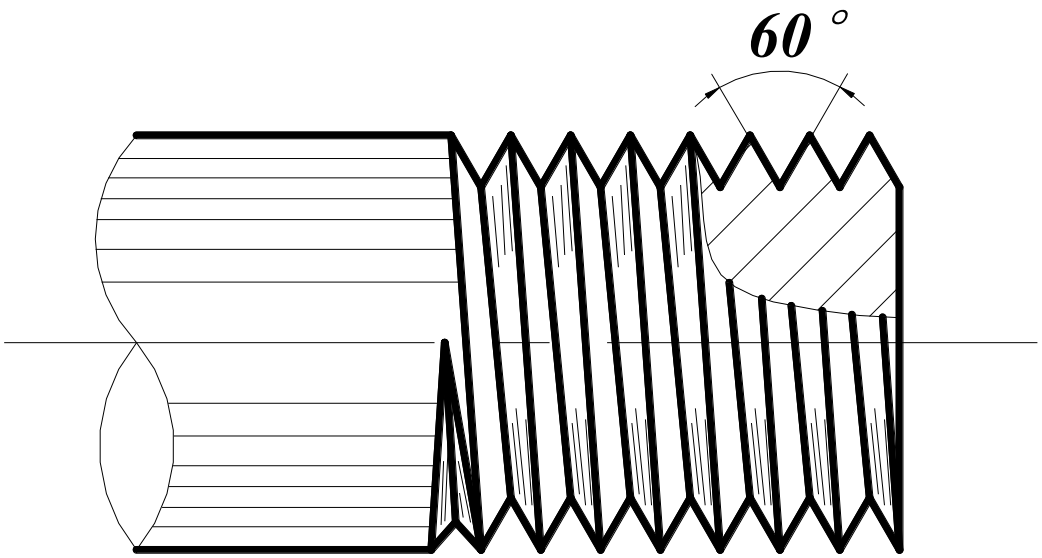
在轴剖面上螺纹的轮廓形状

①普通螺纹 M（M为牙型代号）

牙型：三角形

牙型角：60°

普通螺纹有粗牙、细牙之分

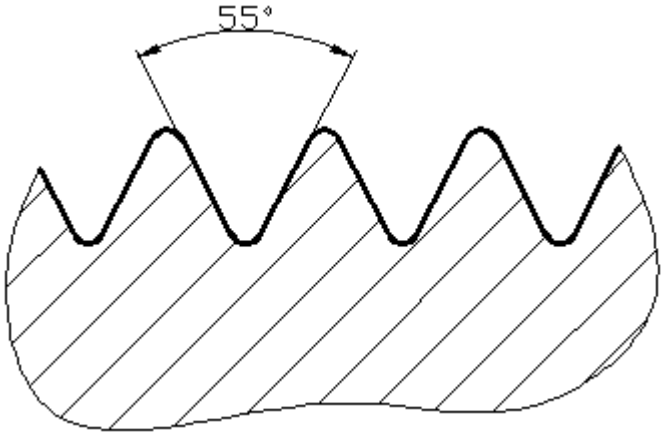


②管螺纹　英制螺纹

管螺纹主要用于连接管子，牙型为三角形，牙型角为55°，管螺纹有两类：

(1) 非螺纹密封的圆柱管螺纹　螺纹种类为*G*，其内、外螺纹均为圆柱螺纹，内外螺纹旋合无密封能力，常用于电线管等不需要密封的管路中的连接。

(2) 螺纹密封的管螺纹　螺纹种类代号有三种：圆锥外螺纹为*R*；圆锥内螺纹（锥度1：16）为R_C；圆柱内螺纹为R_P。这种螺纹可以是圆锥内螺纹与圆锥外螺纹相连接；也可是圆柱内螺纹和圆锥外螺纹相连接，其内、外螺纹旋合后有密封能力，常用于水管、煤气管、润滑油管等。

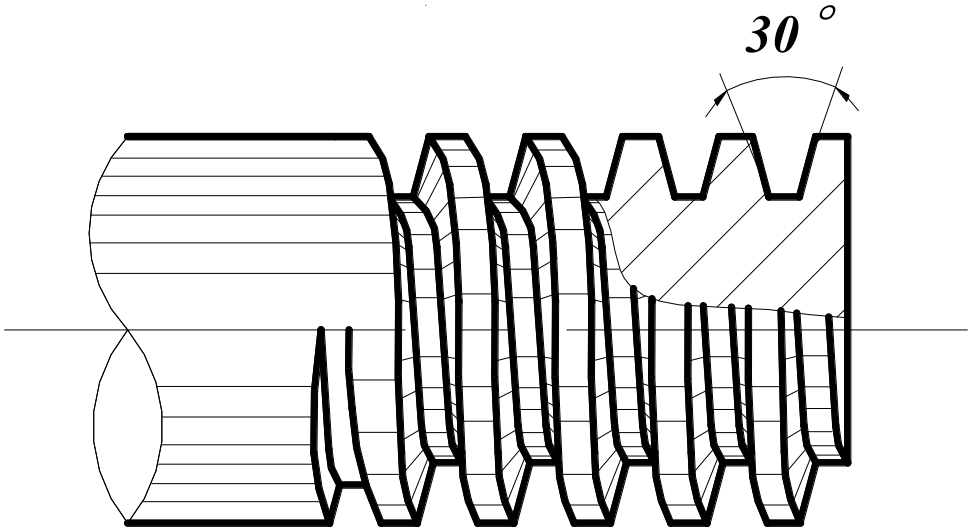


③梯形螺纹Tr

牙型：梯形

牙型角：30°

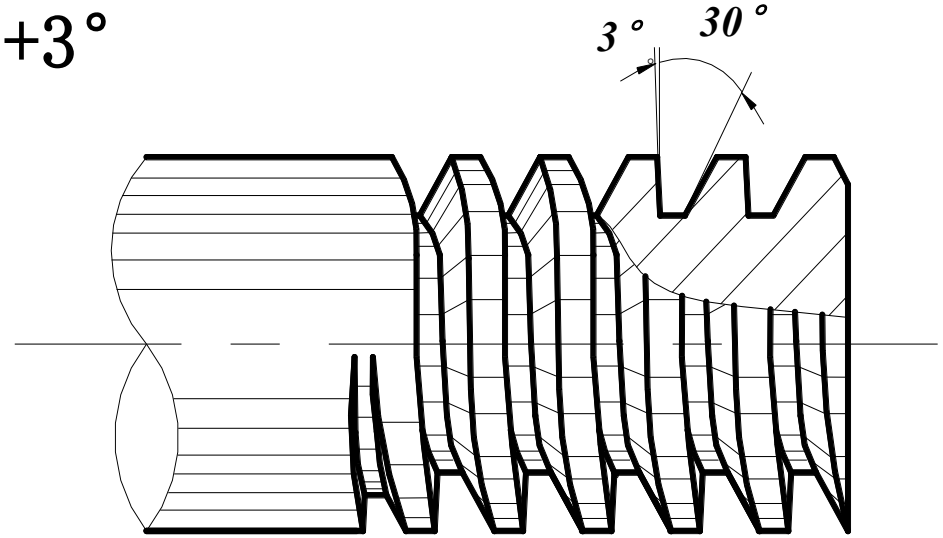
牙顶、牙底都很厚



④锯齿形螺纹B

牙型：锯齿状

牙型角：30° +3°



⑤矩形螺纹

非标准螺纹无代号

♦牙型小结

①普通螺纹M　牙型:三角形　牙型角：60°

②非螺纹密封圆柱管螺纹 G

牙型:三角形 牙型角：55°

③梯形螺纹Tr　牙型:梯形　牙型角：30°

④锯齿形螺纹B　牙型:锯齿状;牙型角：30°+3°

⑤矩形螺纹 无代号　牙型：矩形

(2) 公称直径 （大径）

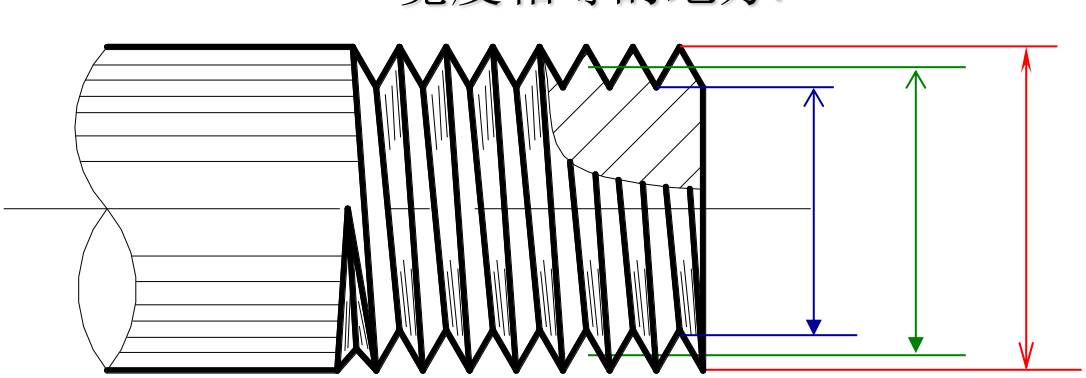
螺纹的直径有大径（d或D）、小径（d_1或D_1）、中径（d_2或D_2）之分。

普通螺纹和梯形螺纹的大径又称公称直径。

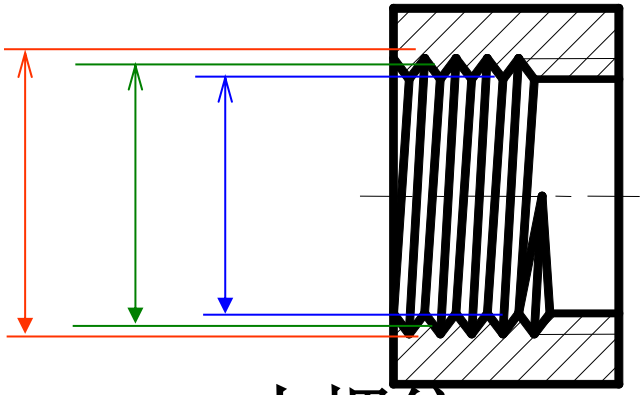
(1) 大径：与外螺纹牙顶或内螺纹牙底相重合的假想圆柱面的直径称为大径。

(2) 小径：与外螺纹牙底或内螺纹牙顶相重合的假想圆柱面的直径称为小径。

(3) 中径：是个假想圆柱的直径。此圆柱的母线通过牙型上沟槽和凸起宽度相等的地方.



外螺纹

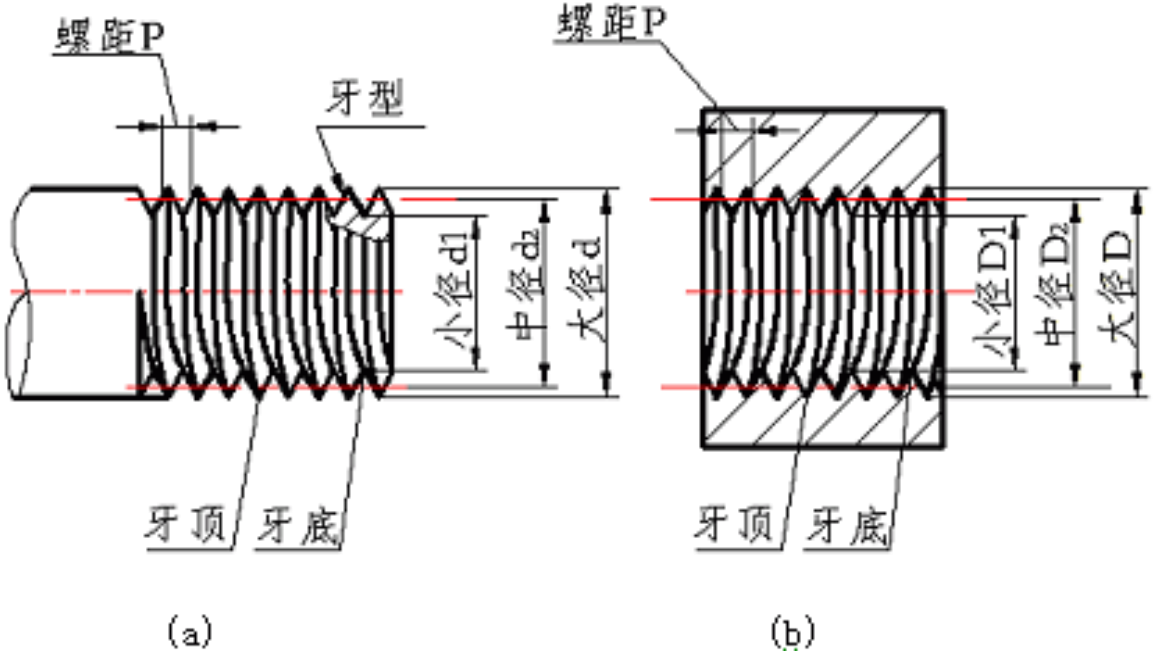


内螺纹

(3) 螺距P

在轴剖面上中径上相邻两个牙对应点之间的轴向距离。

普通螺纹有粗牙和细牙之分：
粗牙螺距标准，细牙螺距不唯一



(4) 线数N与导程S

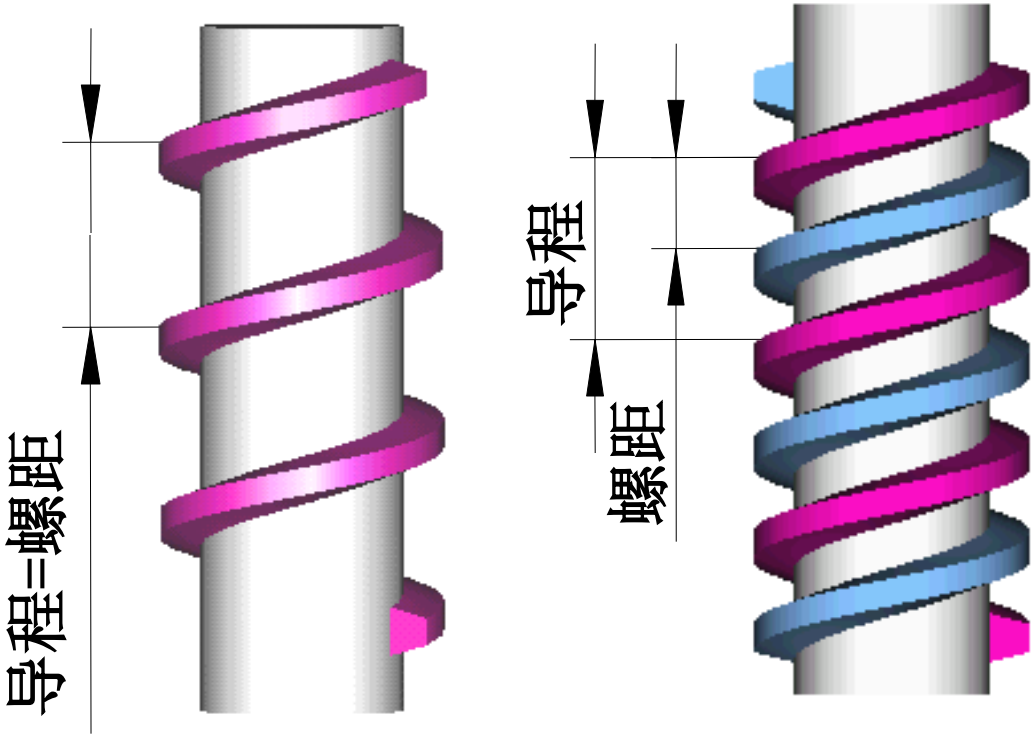
线数：

螺旋线有几条

导程：

同一条螺旋线中经上，相邻两牙间的轴向距离。

$S=N\times P$

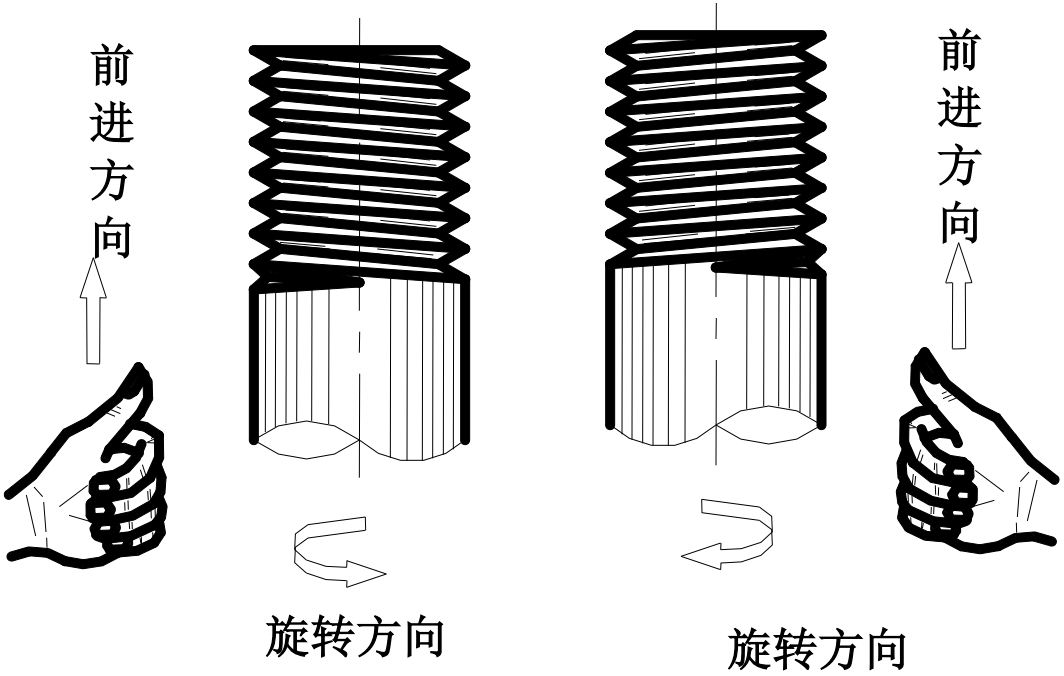


单线螺纹　　双线螺纹

(5) 旋向

螺纹有右旋、左旋之分

如何判定?



(a)左旋　　(b)右旋

螺纹按牙型可分为：

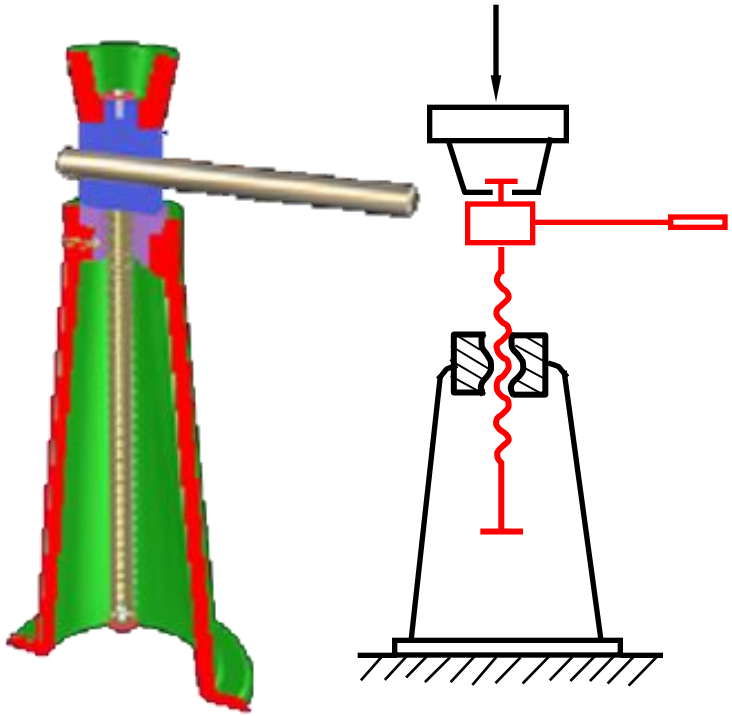
- 按用途分
  - 连接螺纹
    - a. 普通螺纹 M
    - b. 管螺纹 G
  - 传动螺纹
    - c. 梯形螺纹Tr
    - d. 锯齿形螺纹B
    - e. 矩形螺纹

传动螺纹例子

千斤顶

汽车换胎用的

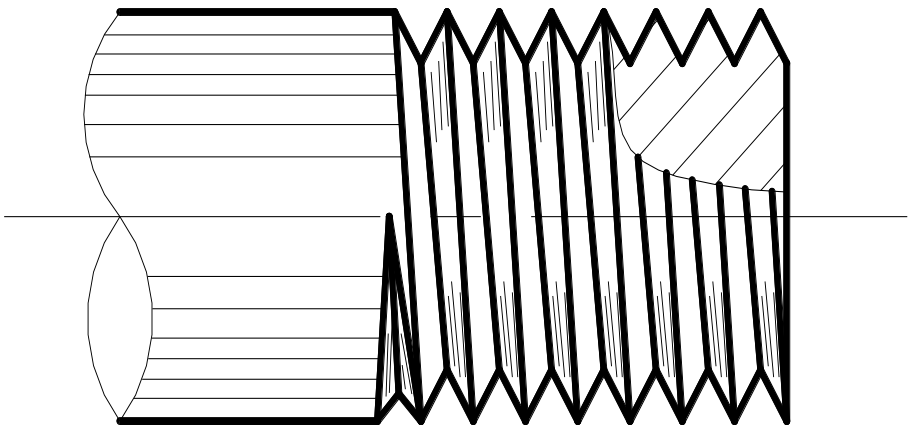
矩形螺纹



四、螺纹的画法

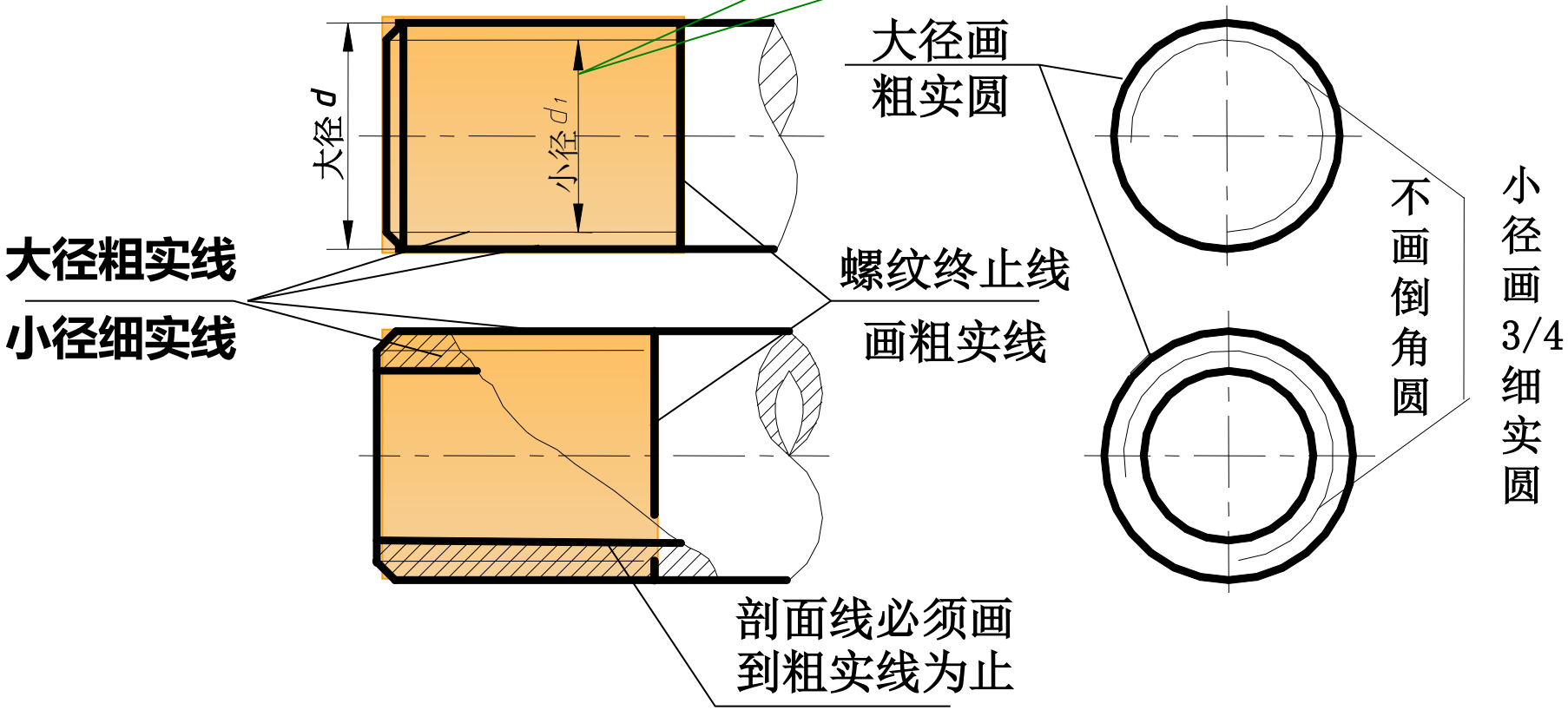
如何来画螺纹?规定的简化画法

外螺纹\内螺纹\螺纹的旋合画法

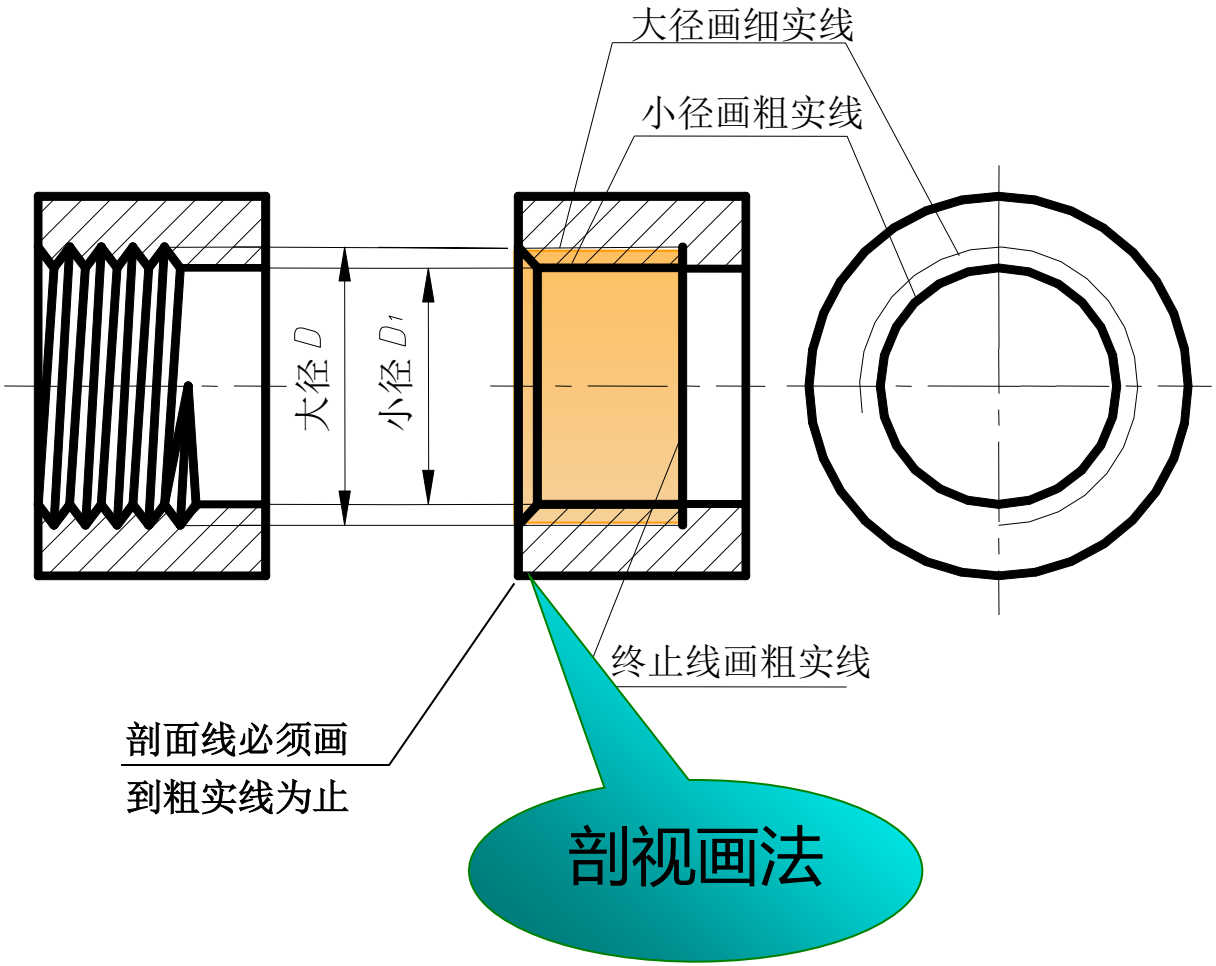


1.外螺纹的画法

一般近似地取小径为大径的0.85倍

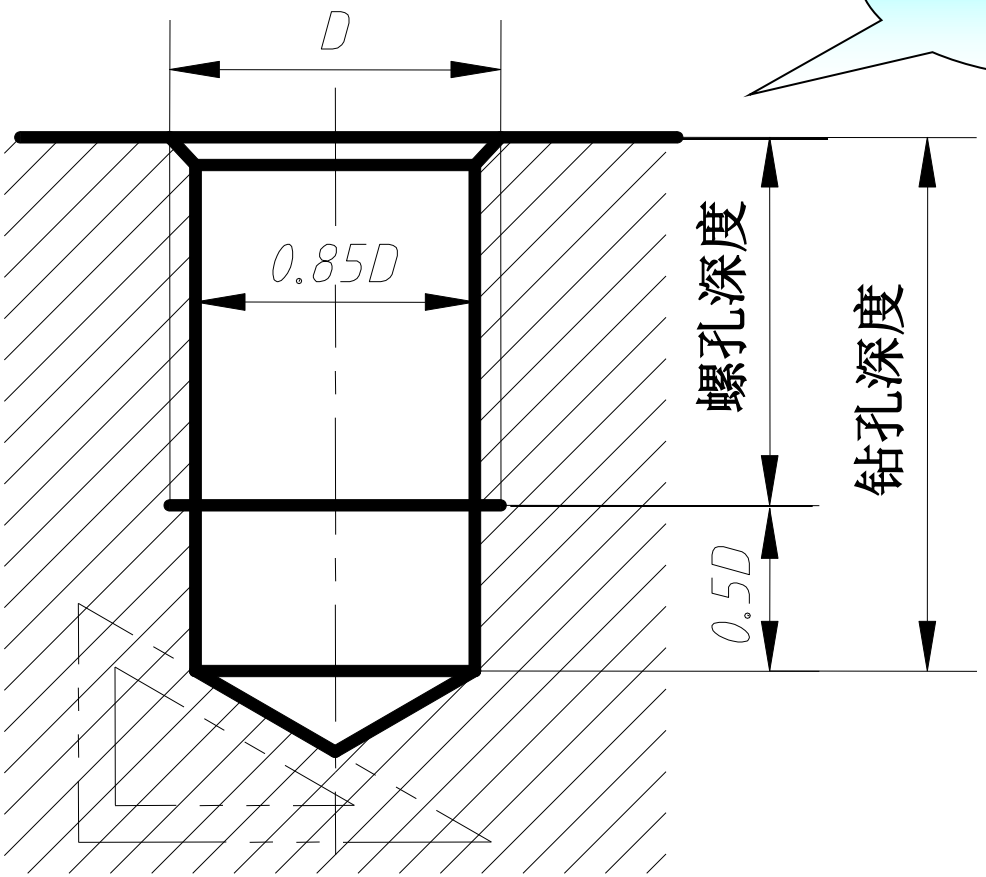


2. 内螺纹的画法



§7-1 螺纹

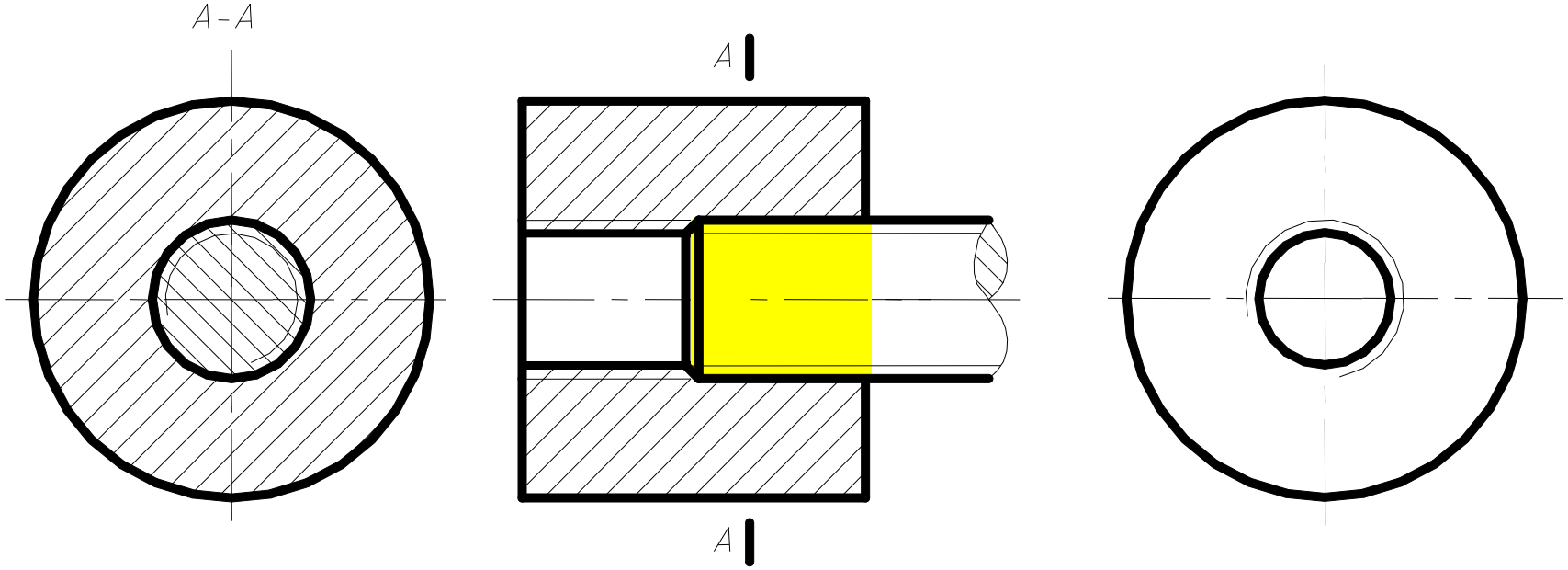
不通孔的画法



1. 倒角
2. 钻孔
3. 加工螺纹

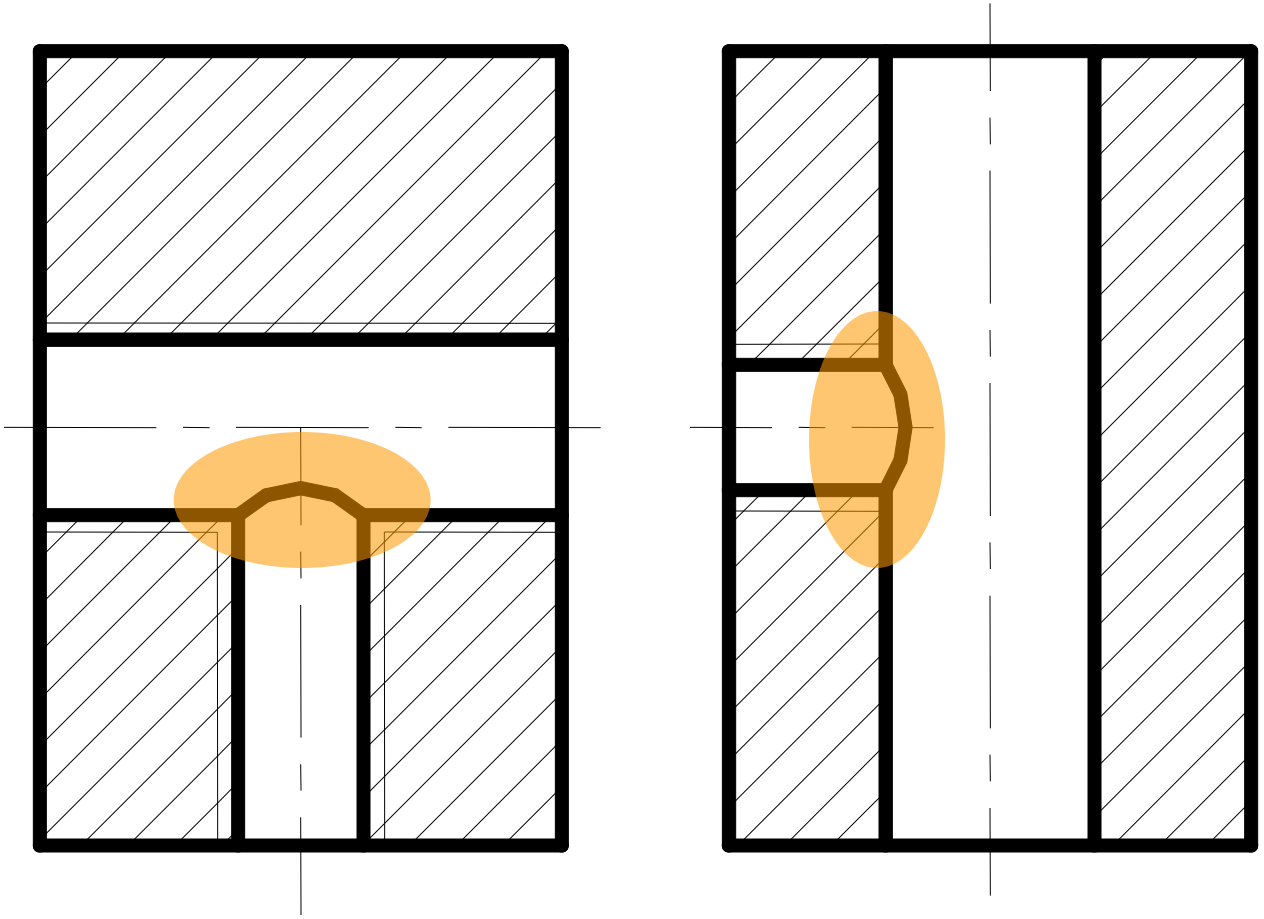
3. 内外螺纹旋合的画法

旋合部分按外螺纹画法， 其余部分按各自画法。



注意:此图是小小装配图, 应按照装配图的规定来画。

4. 螺纹孔相贯线的画法



五、螺纹的标注

螺纹完整标注格式如下：

特征代号 | 公称直径 X 导程(P螺距) | 旋向 — 公差带代号 — 旋合长度代号

牙型　　大径

精度 ⇓ s、m、l

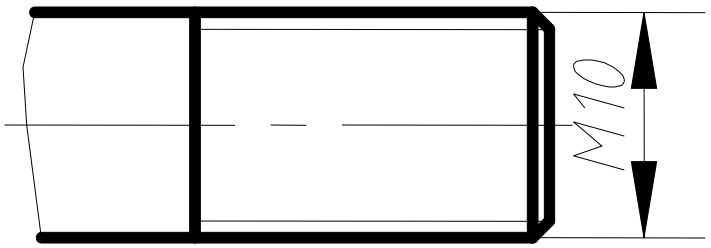
可不标

(1) 普通螺纹的标注

M10

公称直径

特征代号



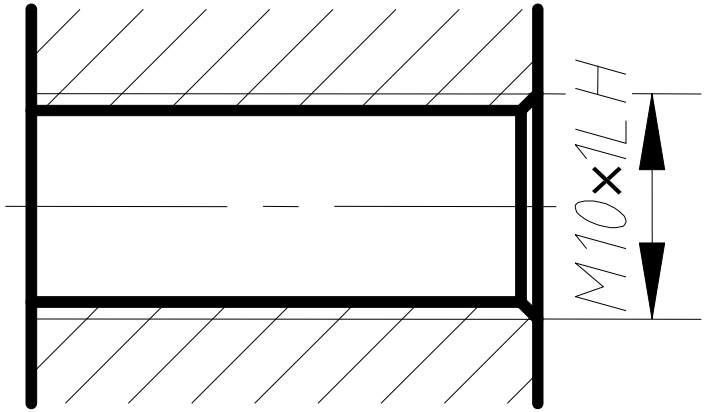
M10 × 1 LH

左旋

螺距

公称直径

特征代号



粗牙螺纹不注螺距，左旋螺纹标注 LH，右旋不标注

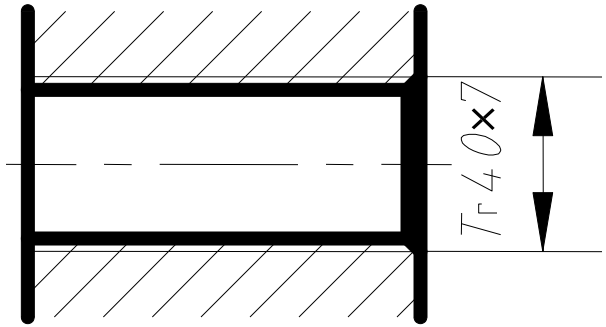
(2) 梯形螺纹的标注

$T_r40\times7$

螺距

公称直径

特征代号



$T_r40\times14(P7)LH$

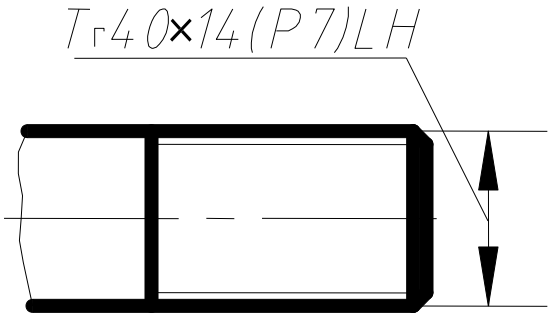
左旋

螺距

导程

公称直径

特征代号



左旋螺纹柱“LH”，

右旋不标注

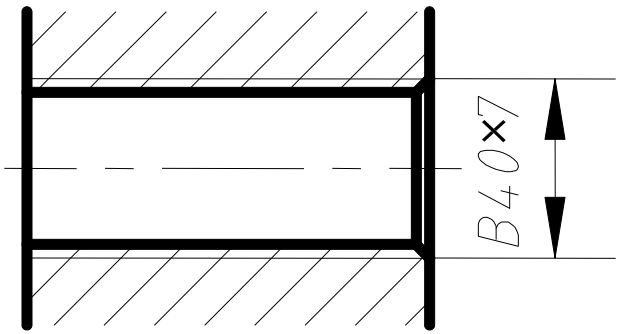
(3) 锯齿形螺纹的标注

B40×7

螺距

公称直径

特征代号



B40 ×14(P7)LH

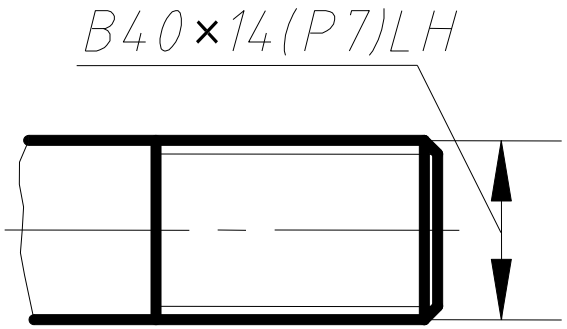
左旋

螺距

导程

公称直径

特征代号

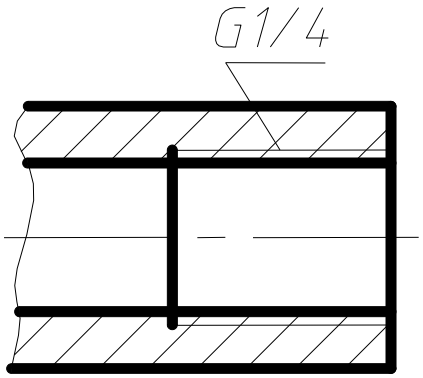


(4) 管螺纹的标注

G1/4

尺寸代号

特征代号



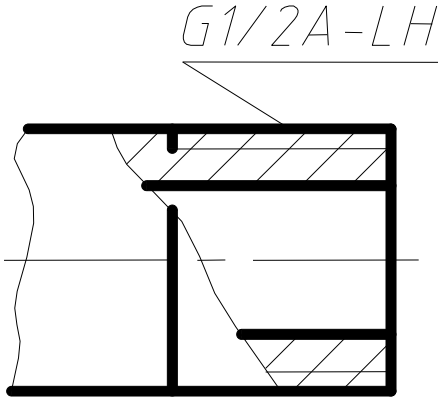
G1/2A-LH

左旋

等级代号

尺寸代号

特征代号

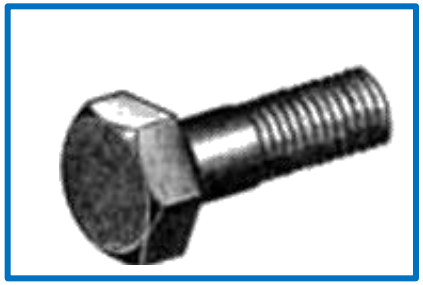


螺纹中径公差分A、B两级

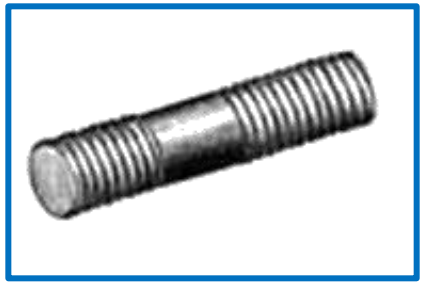
左旋螺纹注“-LH”，右旋不标注

一、常见的螺纹紧固件

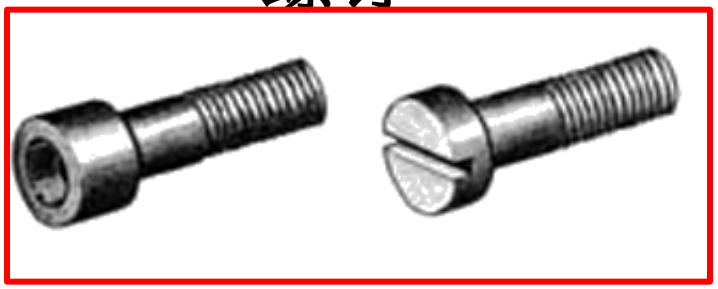
螺钉



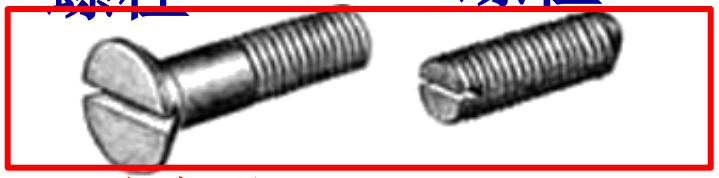
螺栓



螺柱



内六角螺钉　盘头螺钉



沉头螺钉　紧定螺钉

平垫圈

弹簧垫圈

螺母

止口垫圈

二、螺纹紧固件的规定标记

紧固件有完整标记和简化标记两种方法

简化标记如下：

名称 标准代号 规格尺寸(公称直径x公称长度)

【例1】 六角头螺栓，公称直径d＝10mm，公称长度45mm。

螺栓 GB/T 5782-2000 M10x45

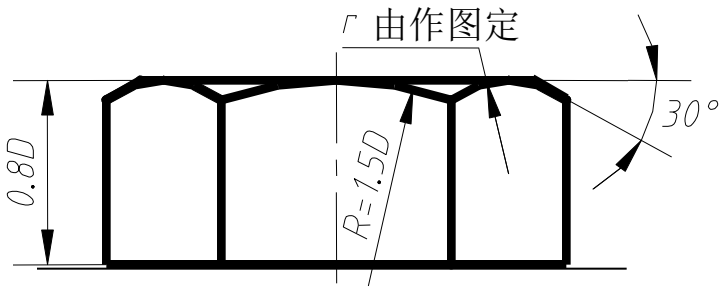
三、螺纹紧固件的规定画法

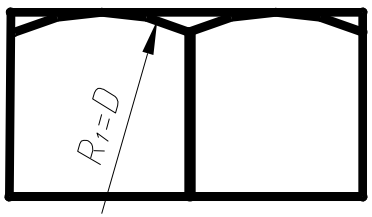
（1） 查表画法：

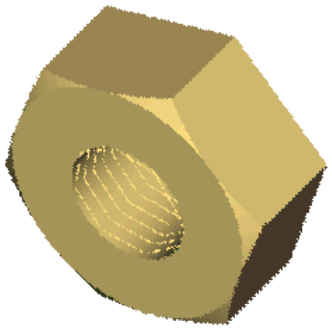
（2） 比例画法：

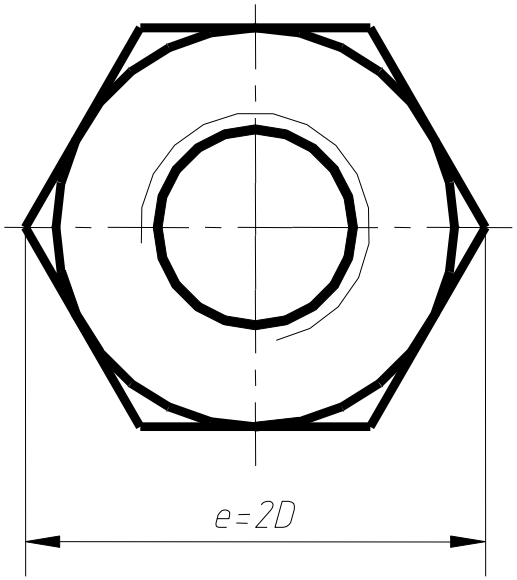
尺寸用公称直径的加权表示

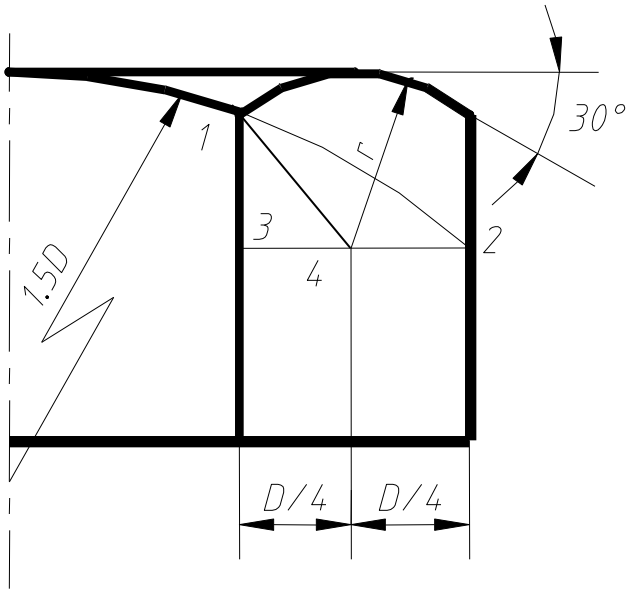
①螺母的画法











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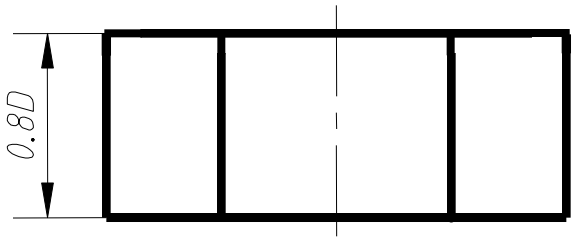
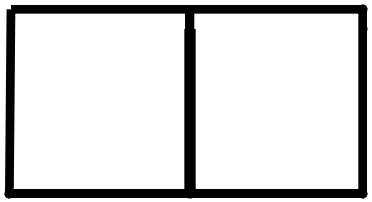
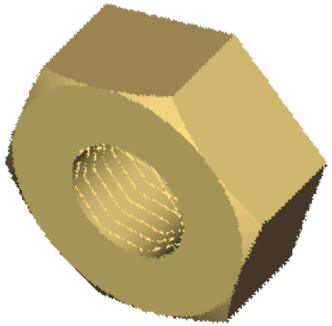
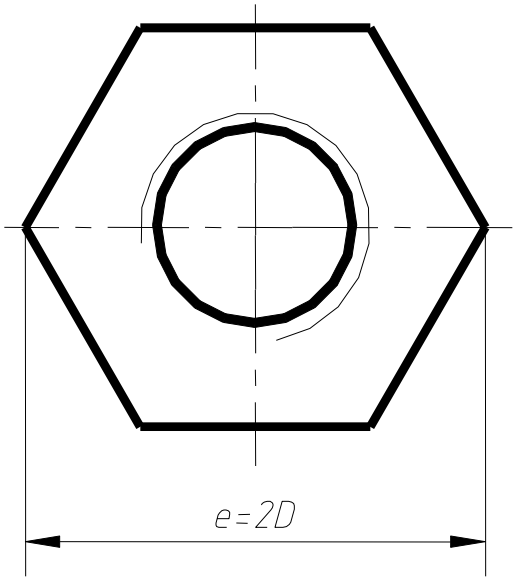
①螺母的画法



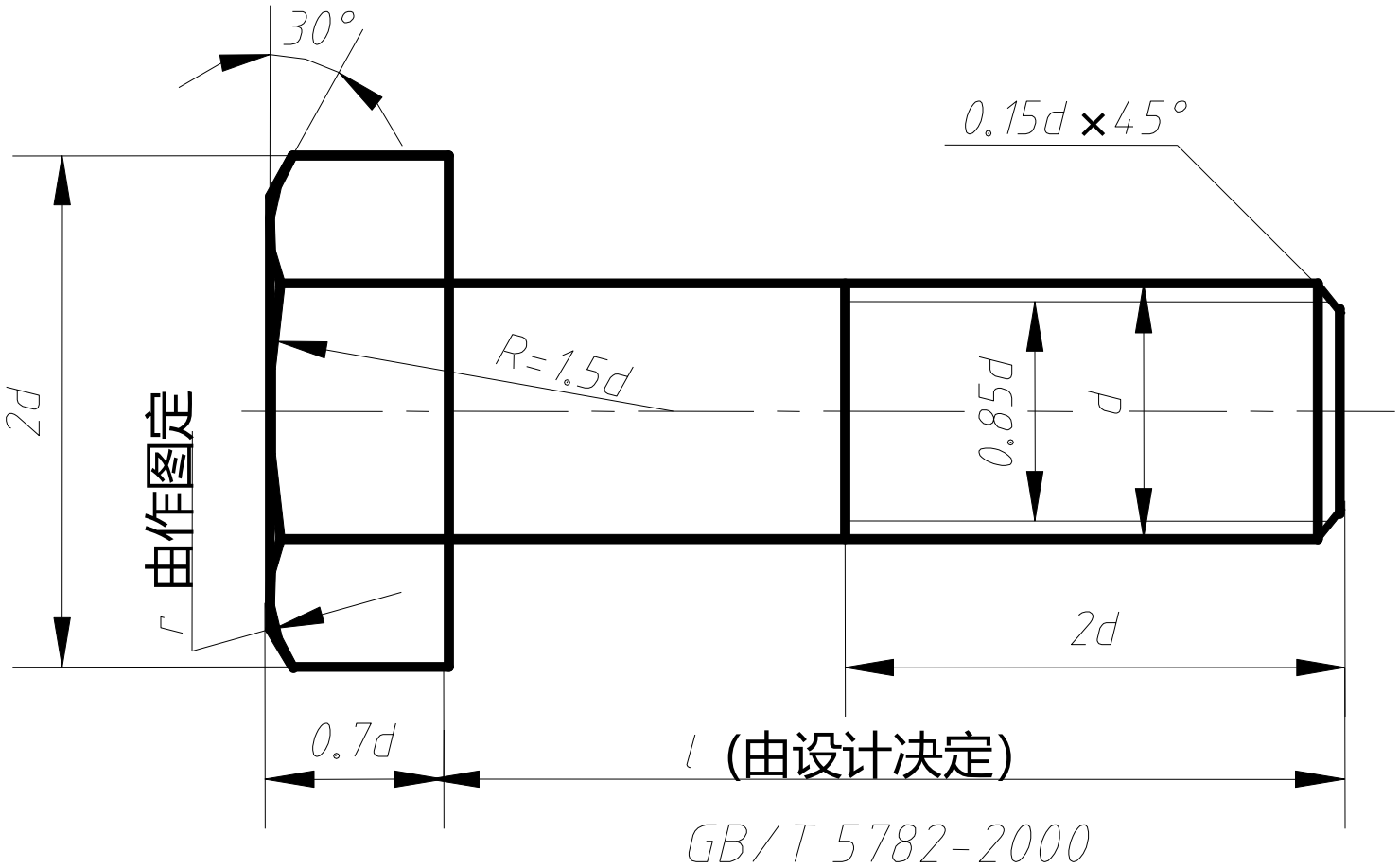






简化画法

②螺栓的画法



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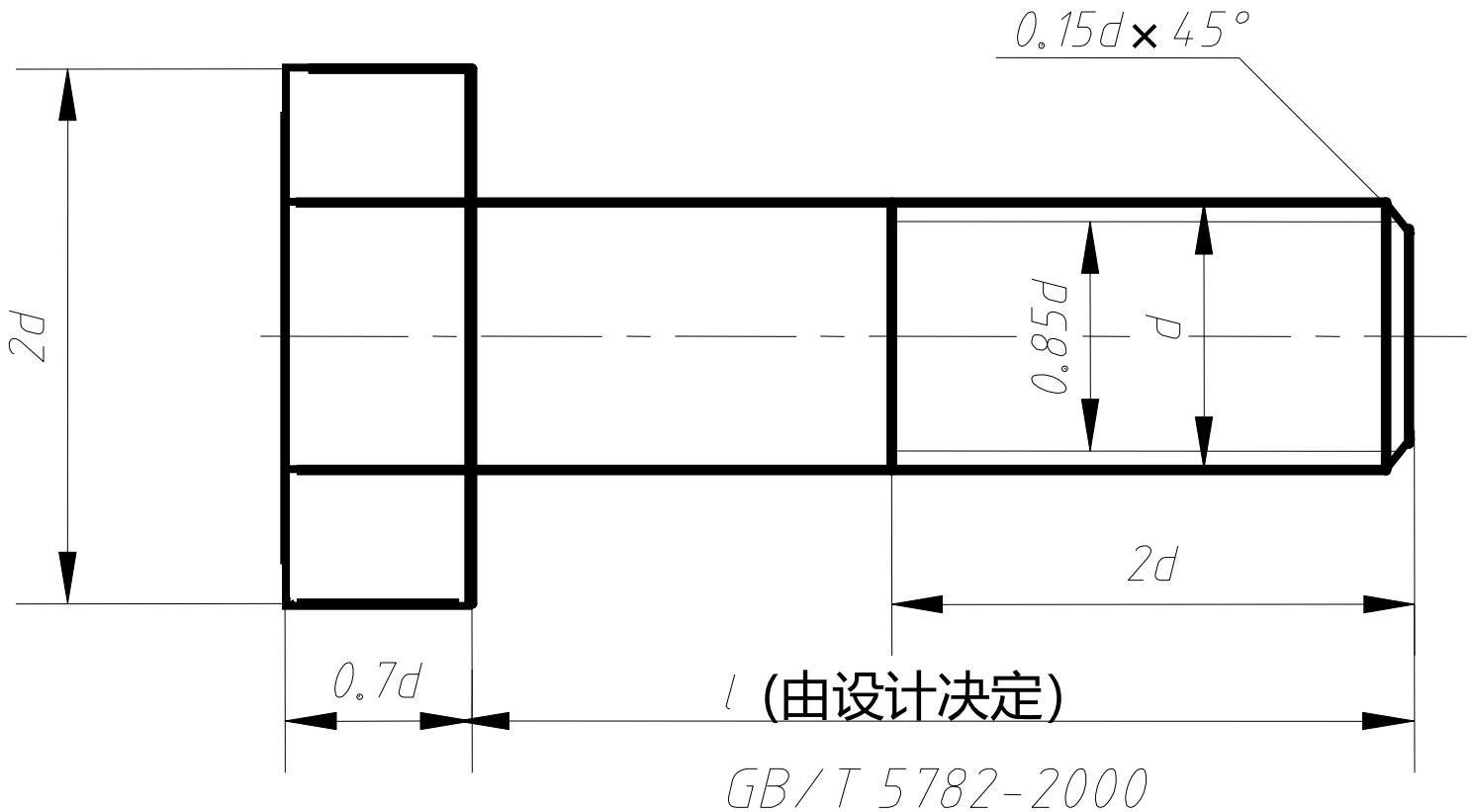
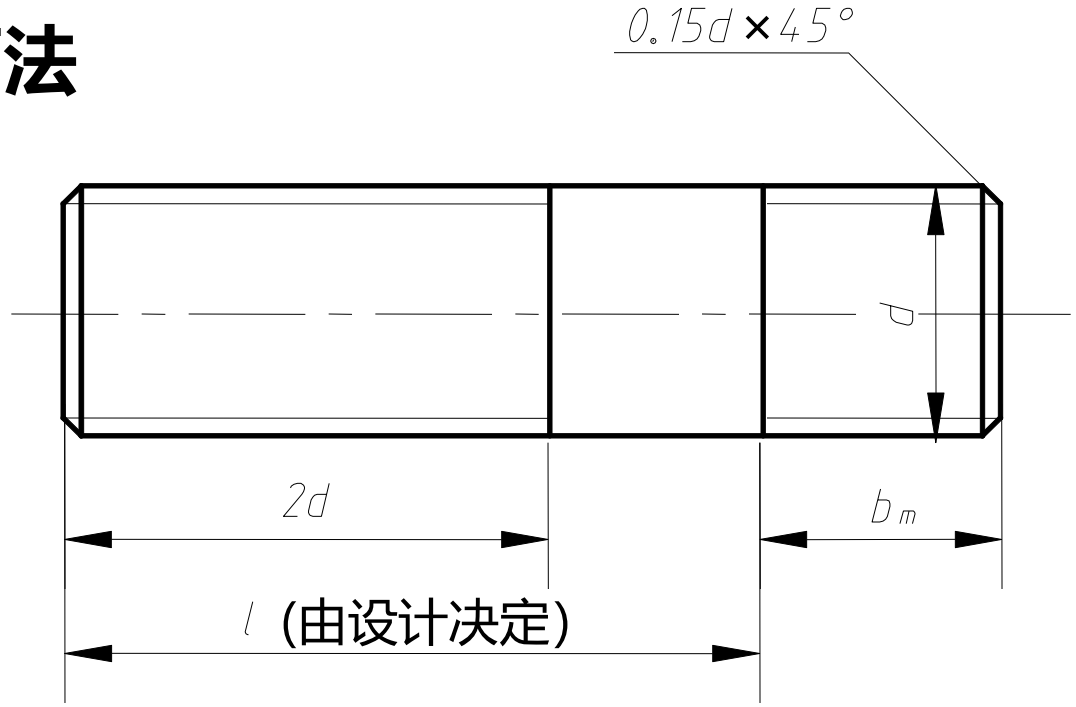
②螺栓的画法

简化画法



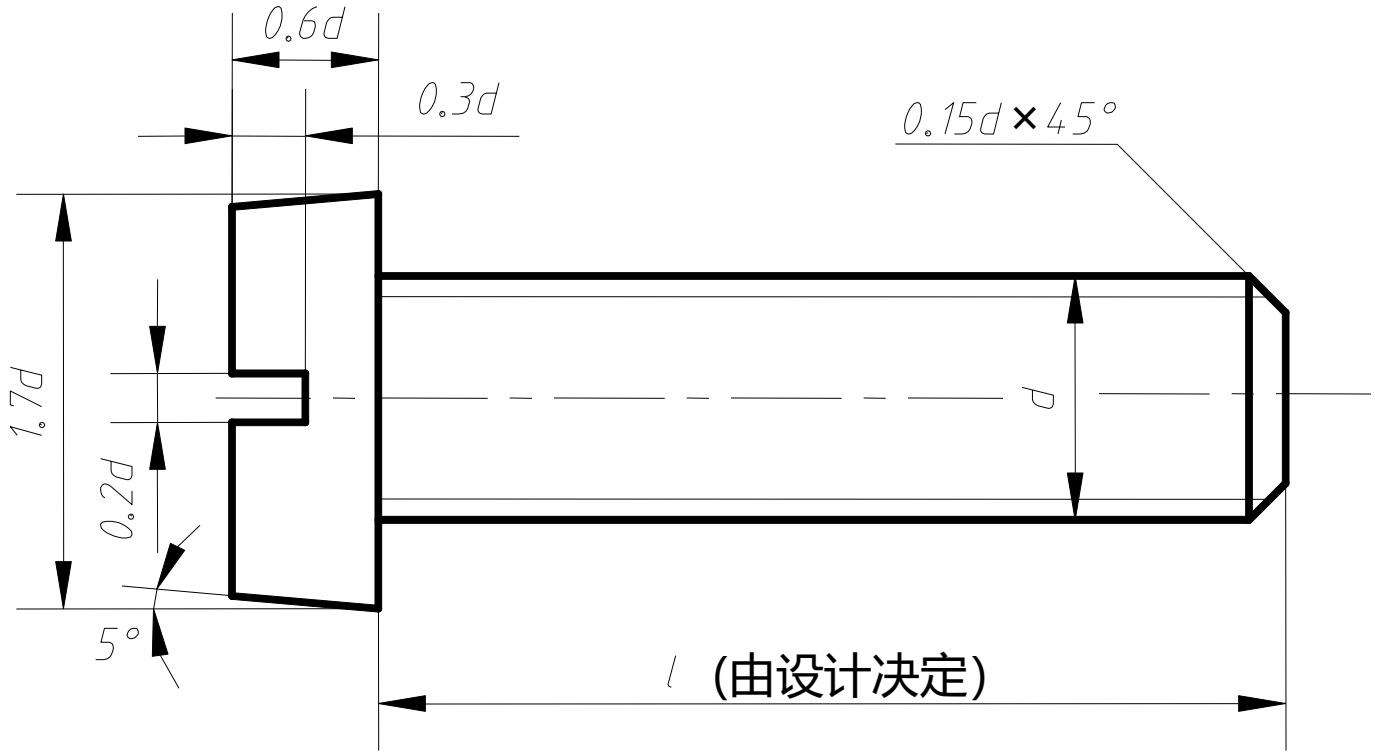
③螺柱的画法

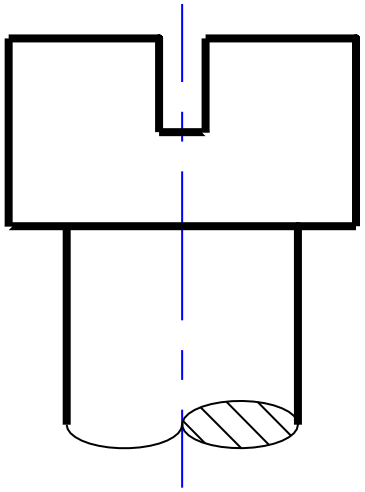


bm为旋入段

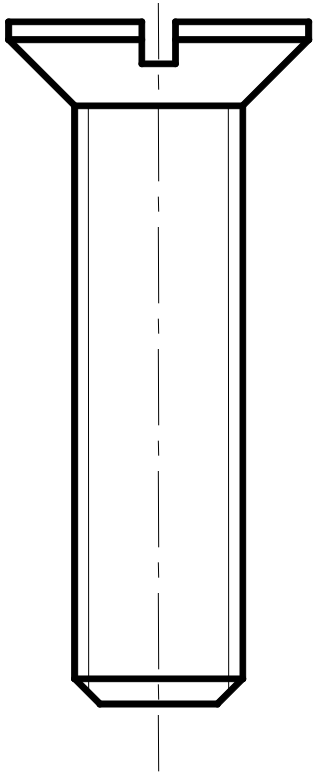
被旋入零件的材料	旋入端长度	国标
钢、青铜	$b_m=d$	GB/T897—1988
铸铁	$b_m=1.25d$	GB/T898—1988
在铸铁和铝之间	$bm=1.5d$	GB/T899—1988
铝	$b_m=2d$	GB/T900—1988

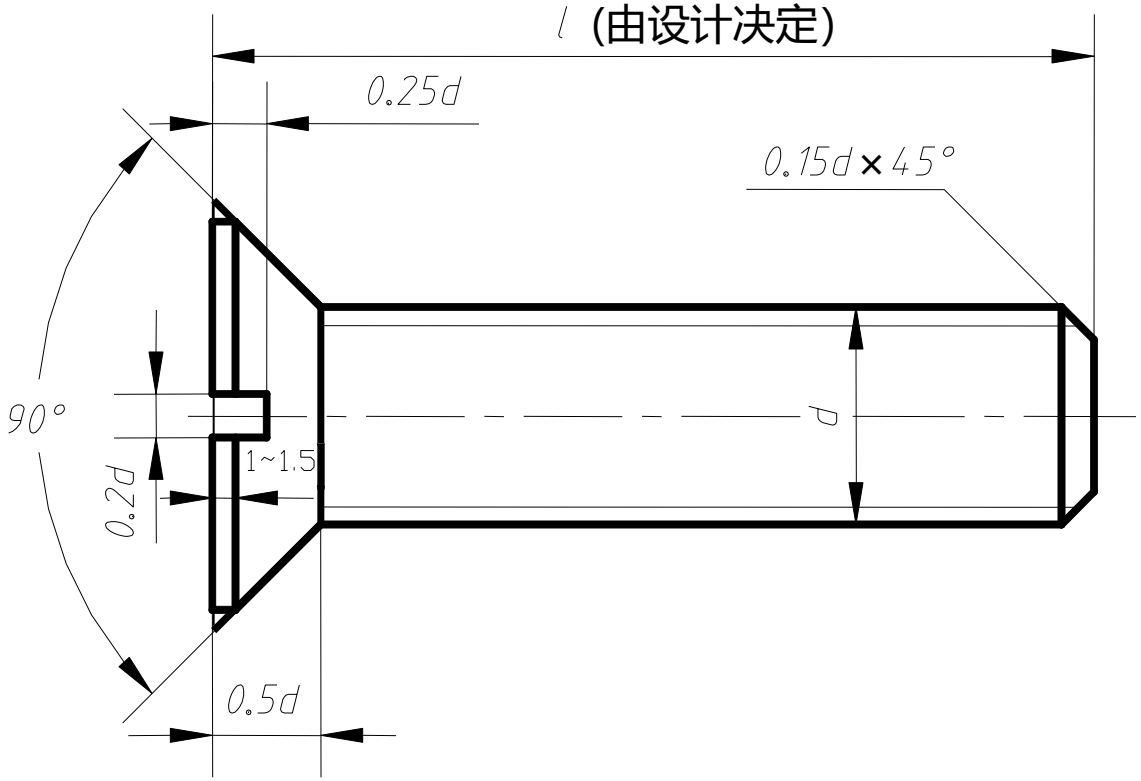
④螺钉头的画法





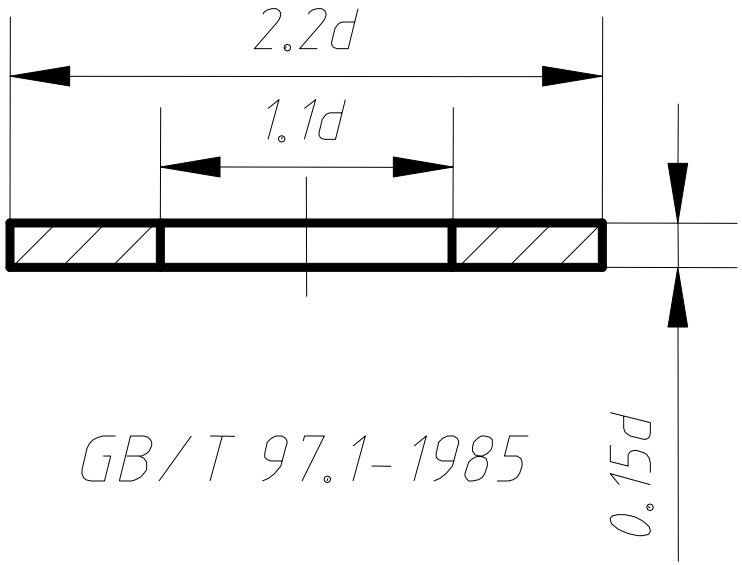
§7-2　螺纹紧固件及其连接





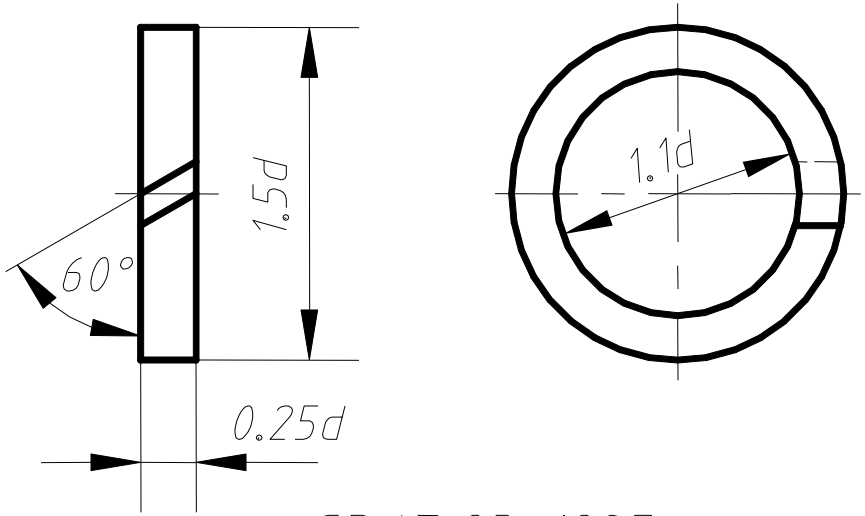
⑤垫圈的画法

平垫圈画法



GB/T 97.1-1985

弹簧垫圈的画法



GB/T 93-1987

注意角度的画法

四、螺纹紧固件的装配图画法

螺纹紧固件装配画法的规定：

（1）两零件的接触面画一条线，否则应画两条线。

（2）相邻两零件的剖面线方向要相反或间隔不同。

（3）当剖切平面通过螺杆的轴线时，螺栓、螺柱、螺钉及螺母、垫圈等均按不剖绘制。

§7-2　螺纹紧固件及其连接

螺栓连接

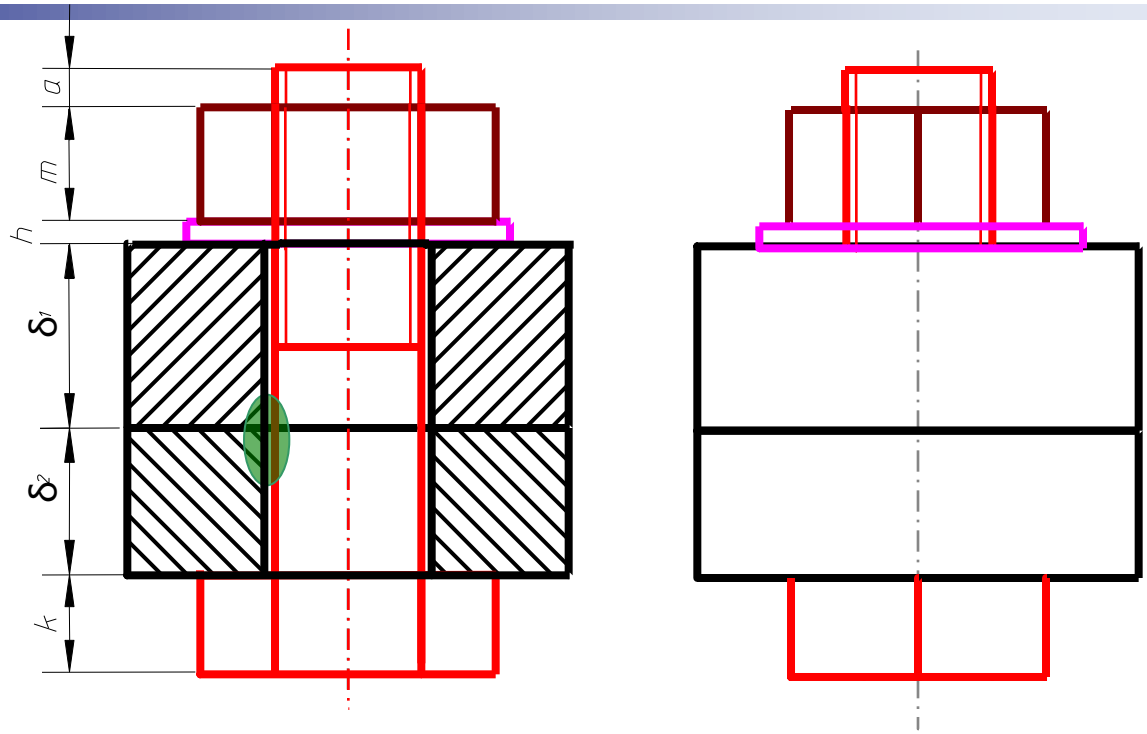
1. 表达：

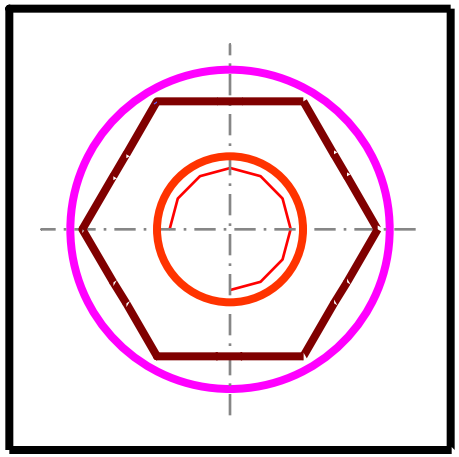
   两块板

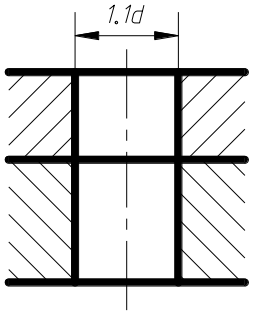
2. 螺栓

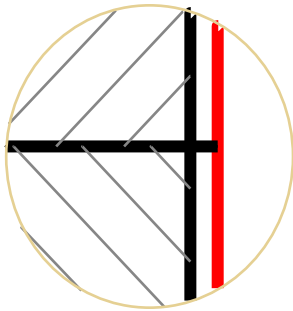
3. 垫圈

4. 螺母









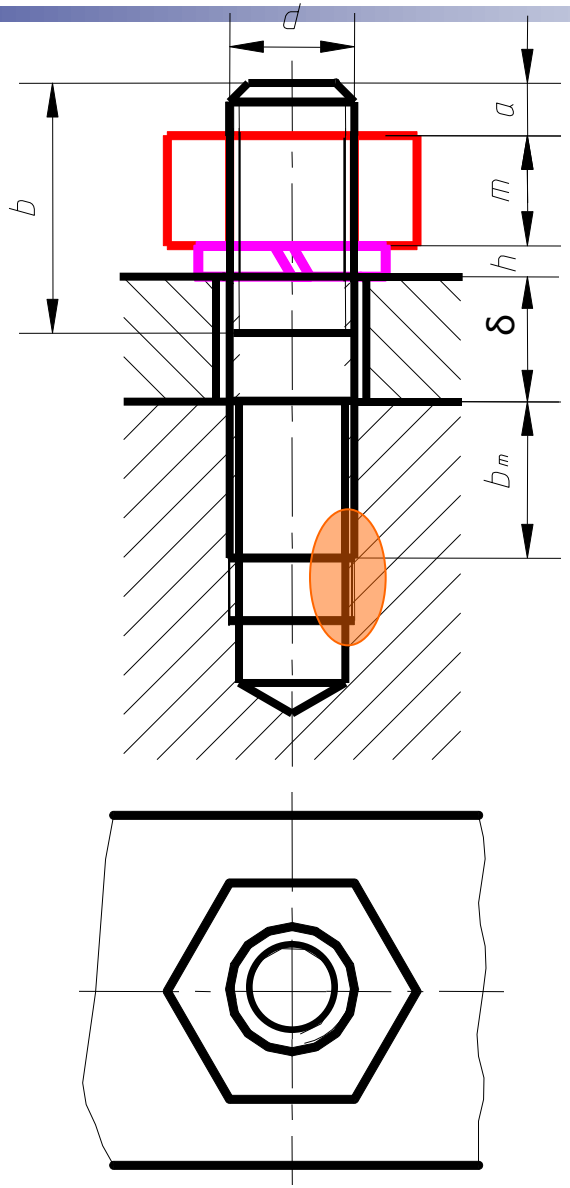
螺柱连接

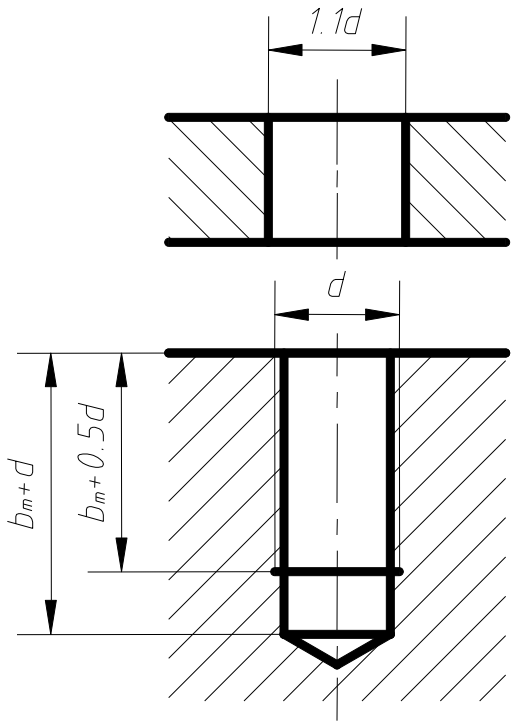
1. 上板、下板

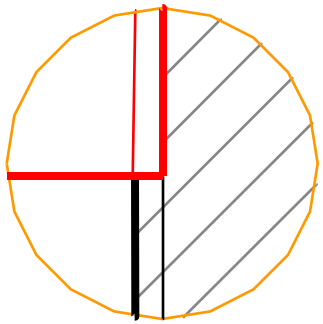
2. 螺柱

3. 垫圈

4. 螺母







螺柱连接

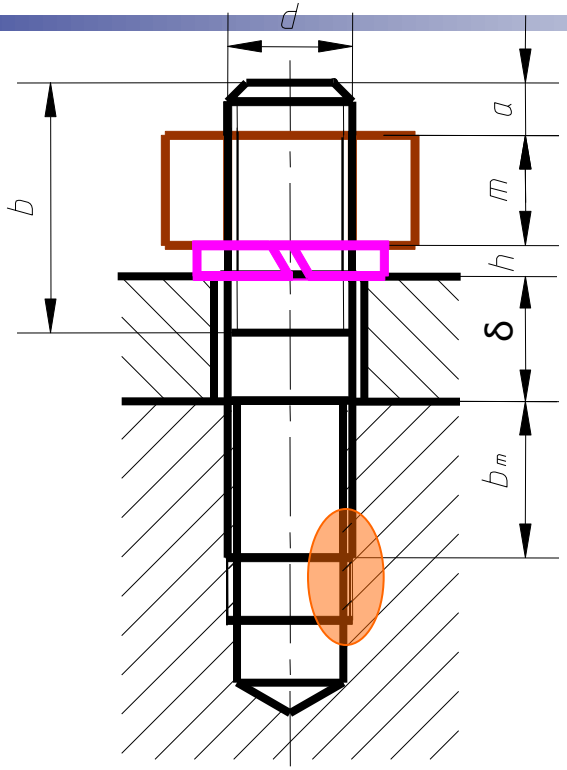
1. 上板薄

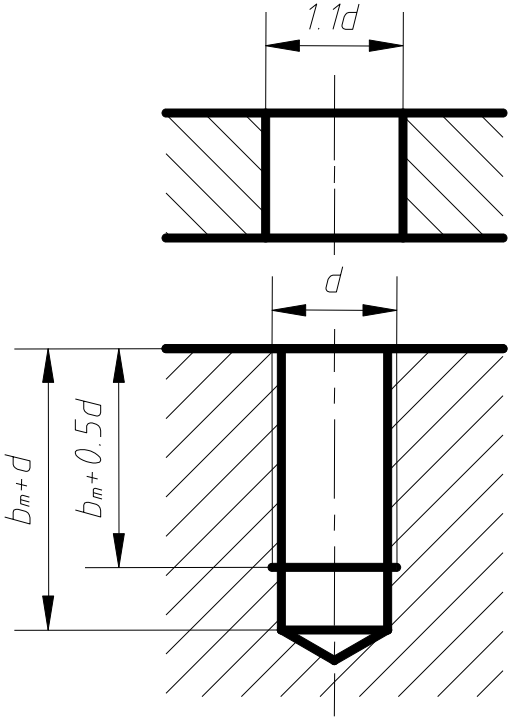
   下板厚

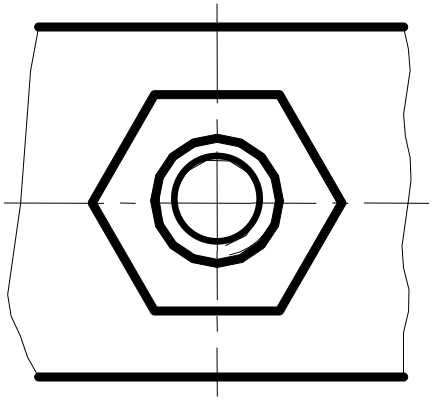
2. 螺柱

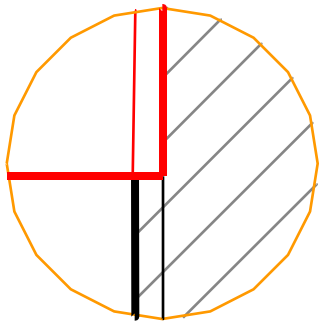
3. 垫圈

4. 螺母









关于旋入端长度：

钢或青铜：bm=d

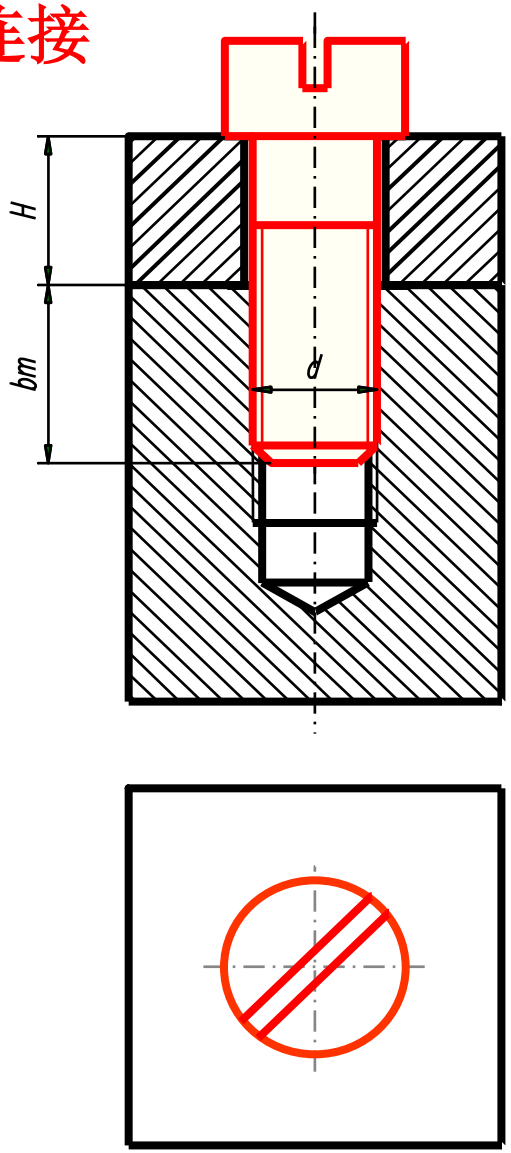
铸铁：bm=1.25d

铝合金：bm=2d

强度在铸铁和铝之间的材料：bm=1.5d

螺钉连接

螺钉连接的简化比例画法



圆柱头螺钉连接的画法

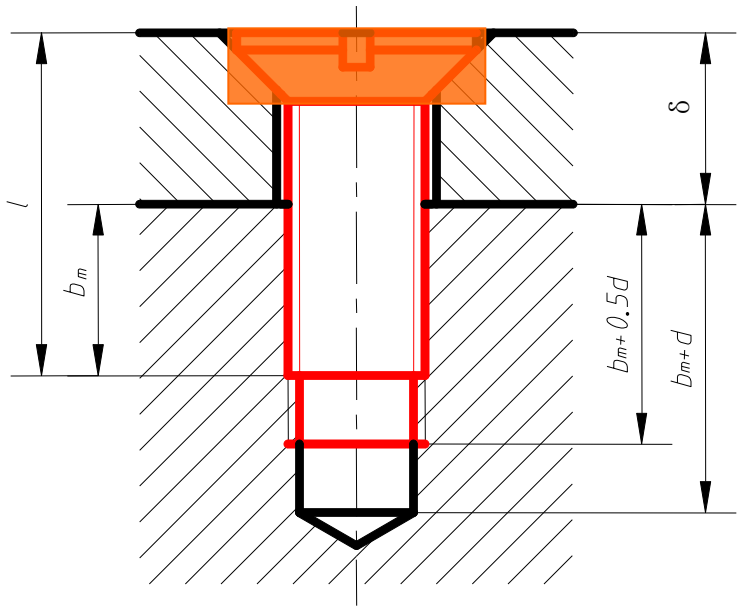
孔径

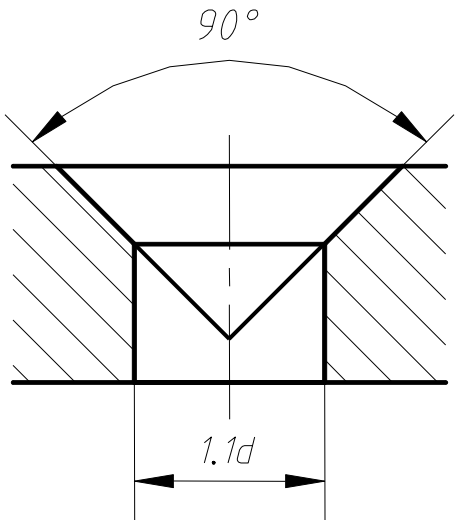
螺纹终止线位置

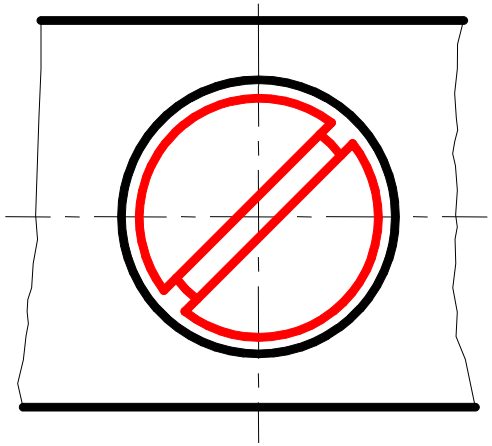
剖面线

俯视图开口方向

螺钉连接的简化比例画法







沉头螺钉连接的画法